

FitBuddy – AI Fitness Plan Generator

Project Development Phase – Page 1

Development Phase Overview

The Development Phase converts the approved design into working software. FitBuddy is developed as connected frontend, backend, AI and database modules.

- Create project folders
- Install dependencies
- Implement functions
- Connect modules
- Run the application

Development Approach

Development is performed module by module and then integrated into one application.

FitBuddy – AI Fitness Plan Generator

Project Development Phase – Page 2

Project Structure Development

The project structure separates application logic, AI generation, database operations, templates and configuration.

- Backend Python files
- AI generator modules
- Database module
- HTML templates
- Environment/configuration files

Dependency Setup

Required packages include FastAPI, Uvicorn, Jinja2, SQLAlchemy, python-multipart and Google Gemini-related package support.

FitBuddy – AI Fitness Plan Generator

Project Development Phase – Page 3

Workout Generation Development

The workout generator receives fitness information and requests a structured 7-day plan from the Gemini service.

- Accept goal
- Accept intensity
- Use age and weight context
- Generate 7-day plan
- Return result to backend

Nutrition Development

A separate function generates nutrition or recovery guidance so this responsibility remains independent from workout generation.

FitBuddy – AI Fitness Plan Generator

Project Development Phase – Page 4

Database Development

Database functions are developed for storing users and plans and for retrieving or updating plan information.

- save_user()
- save_plan()
- update_plan()
- get_original_plan()
- get_user()

Data Persistence

Generated plans are stored so that the original plan and later updated plan can be accessed by the application.

FitBuddy – AI Fitness Plan Generator

Project Development Phase – Page 5

Route Development

FastAPI routes connect the user interface with the application logic.

- / – input page
- /generate-workout – generation
- /submit-feedback – update
- /view-all-users – stored users

Request Processing

Routes receive form data, call the required functions, save results and return the appropriate HTML template.

FitBuddy – AI Fitness Plan Generator

Project Development Phase – Page 6

Frontend Development

The frontend is developed using HTML/Jinja2 templates for input, result display and the all-users view.

- index.html
- result.html
- all_users.html
- Forms and result sections
- Readable layout

User Experience

The interface is designed to be simple and responsive so users can enter information and understand the generated plan easily.

FitBuddy – AI Fitness Plan Generator

Project Development Phase – Page 7

Feedback Update Development

The feedback module passes the original plan and user feedback to the update function, which creates a revised plan.

- Receive feedback
- Retrieve original plan
- Generate updated plan
- Store updated plan
- Display updated result

Integration Development

The AI, database, routes and templates are connected and tested together as a complete workflow.

FitBuddy – AI Fitness Plan Generator

Project Development Phase – Page 8

Deployment Development

The application is prepared to run using Uvicorn with environment-based API configuration.

- Create virtual environment
- Install dependencies
- Configure API key
- Start server
- Test deployed/local flow

Development Quality Checks

Code is checked for import errors, route errors, missing templates, database issues and AI configuration problems.

FitBuddy – AI Fitness Plan Generator

Project Development Phase – Page 9

Final Development Summary

The Development Phase produces the working FitBuddy application from the approved design.

- Backend implemented
- AI integration completed
- Database connected
- Frontend connected
- Feedback updating implemented
- Application ready for testing

Development Conclusion

Successful development provides the software foundation required for formal testing and documentation.